Swinburne University of Technology Hawthorn Campus
Department of Computing Technologies

COS20028 Big Data Architecture and Application
Assignment 2 - *Semester 2, 2025*

Assessment Title: Big Data Analytics.
Assessment Outcome: 20%
Due Date: AEST 23:59 on 24/10/2025.
Assessable Item:

- One (1) piece of a report for answers to the given questions.
- One (1) document of the code collections for your assignment.

Purpose of Assignment

The purpose of this assignment is to provide hands-on experience in analysing semi-structured customer loyalty data using Apache Pig. Through a series of practical tasks, students will learn how to load and validate raw text files, transform compound fields into structured attributes, and perform common analytical operations such as grouping, aggregation, filtering, and sorting. The assignment also emphasises deriving business insights, including customer segmentation by membership tier, identifying high-value customers, and exploring spending behaviour. By completing these tasks, students will develop the skills needed to clean, manipulate, and analyse large-scale datasets, while understanding how data processing with Pig can support real-world decision-making in areas such as marketing, customer engagement, and business strategy.

Background

You are given a dataset "loyalty_data.txt" containing loyalty program member information. The source file is located on the path "~/training_materials/analyst/exercises/data_mgmt". Each row is pipe-delimited (|) and includes:

1. **Customer ID**
2. **First Name**
3. **Last Name**
4. **Email**
5. **Membership Tier** (SILVER, GOLD, PLATINUM)
6. **Phone Numbers**
7. **Transactions (comma-separated)**
8. **Account Summary** (comma-separated values: points earned, total spend, purchases, lifetime value)

Assignment Task

Use **only** Apache Pig to solve the given tasks. Your codes must be used to find answers to the questions. Some questions may require multiple steps of code to find answers. Follow the question sequence to build your code. If the answer is found based on the outcome from previous questions, you can provide the code based on the current system state. There is no need to post all the code from the beginning unless the answer cannot be found based on previous process results.

1. Load the source data and find how many tuples are included in this dataset. **[10 marks]**
2. Calculate how many customers belong to each membership tier (SILVER, GOLD, PLATINUM). **[10 marks]**
3. Split the account_summary field into meaningful columns: points, spend, purchases, and lifetime_value. **[10 marks]**
4. Find the top 5 customers with the highest lifetime value. **[10 marks]**
5. Calculate the average total_spend for each membership tier. **[15 marks]**
6. Find customers who have more than one phone number recorded. **[15 marks]**
7. Sum the total points earned by customers in each membership tier. **[15 marks]**
8. Count how many customers use each email domain (e.g., gmail.com, example.com). **[15 marks]**

Please use the template to prepare your submission.

Submission Requirement

Only submissions using the provided report template are acceptable. The submitted code should be pasted in sequence with the plant texts, but not images.
Failure to adhere to the submission requirements will result in immediate failure of this assignment.

Rubric: Assignment 1

	Task	Mark
1.	Task 1	10 marks
2.	Task 2	10 marks
3.	Task 3	10 marks
4.	Task 4	10 marks
5.	Task 5	15 marks
6.	Task 6	15 marks
7.	Task 7	15 marks
8.	Task 8	15 marks

-- End of Assignment --